Chapter 15, Problem 9PP

Problem

Find $f(t)$ if

$$F(s) = \frac{48(s+2)}{(s+1)(s+3)(s+4)}$$

Step-by-step solution

Step 1 of 4

Given $F(s) = \frac{48(s+2)}{(s+1)(s+3)(s+4)}$

Use partial fraction expansion to break $F(s)$ into simple terms

$$\frac{48(s+2)}{(s+1)(s+3)(s+4)} = \frac{A}{s+1} + \frac{B}{s+3} + \frac{C}{s+4} \quad \ldots\ldots (1)$$

Multiplying both sides by $(s+1)(s+3)(s+4)$

$$48(s+2) = A(s+3)(s+4) + B(s+1)(s+4) + C(s+1)(s+3)$$

Consider $s = -1$

$$48(-1+2) = A(-1+3)(-1+4)$$

$$48 = A(2)(3)$$

$$A = 8$$

Step 2 of 4

Consider $s = -3$

$$48(-3+2) = B(-3+1)(-3+4)$$

$$48(-1) = B(-2)(1)$$

$$B = 24$$

Step 3 of 4

Consider $s = -4$

$$48(-4+2) = C(-4+1)(-4+3)$$

$$48(-2) = C(-3)(-1)$$

$$C = -32$$

Step 4 of 4

Thus,

$A = 8,\ B = 24,\ C = -32$

Substitute 8 for A, 24 for B and -32 for C in equation (1).

$$F(s) = \frac{8}{s+1} + \frac{24}{s+3} - \frac{32}{s+4}$$

Determine the inverse Laplace transform of each term.

$$f(t) = L^{-1}\{F(s)\}$$

$$f(t) = L^{-1}\left\{\frac{8}{(s+1)}\right\} + L^{-1}\left\{\frac{24}{(s+3)}\right\} + L^{-1}\left\{\frac{-32}{(s+4)}\right\}$$

$$f(t) = 8L^{-1}\left\{\frac{1}{(s+1)}\right\} + 24L^{-1}\left\{\frac{1}{(s+3)}\right\} - 32L^{-1}\left\{\frac{1}{(s+4)}\right\} \quad \ldots\ldots (2)$$

Use linearity property,

$$L^{-1}\left\{\frac{1}{s+a}\right\} = e^{-at}u(t)$$

Now, apply the linearity property in equation (2).

$$f(t) = 8e^{-t}u(t) + 24e^{-3t}u(t) - 32e^{-4t}u(t)$$

Therefore, $\boxed{f(t) = \left(8e^{-t} + 24e^{-3t} - 32e^{-4t}\right)u(t)}$.